



Heat Transfer

Partial Diff Eq

Heat Equation

- Heat
- Diffusion
- Finance
- Fluid Mech

Computational Fluid Dynamics

Physics	Math	Computational
• Conduction • Convection • Radiation	H.E Diff. Eq	Finite Element method

Comsol multiphysics

FE_{ni}CS

$$\frac{d^2 x}{dt^2} = -kx$$

$$\text{let } x = \sin(kt)$$

$$\frac{dx}{dt} = k \cos(kt)$$

$$\frac{d^2 x}{dt^2} = -k^2 \sin(kt)$$

$$x(t) = A \sin(kt)$$

→ Analytical

↳ This usually is not the direct answer!

x	x	x	x
x	x	x	x
x	x	x	x

→ Numerical Analysis

more datapoints = more accuracy = more computation required!

Electromagnetism

→ for a single variable func: $\frac{d}{dn} f(n) = \frac{d}{dn} f(n)$

$$\frac{d}{dn} f(n,t) \neq \frac{\partial}{\partial n} f(n,t)$$

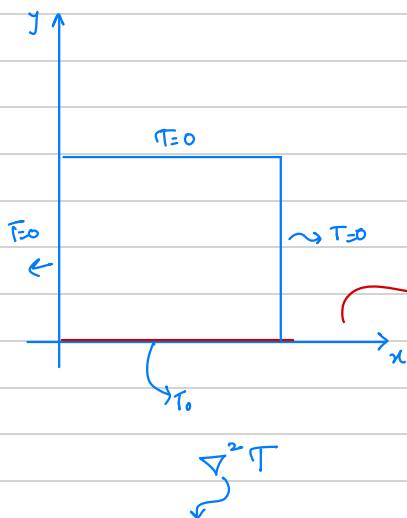
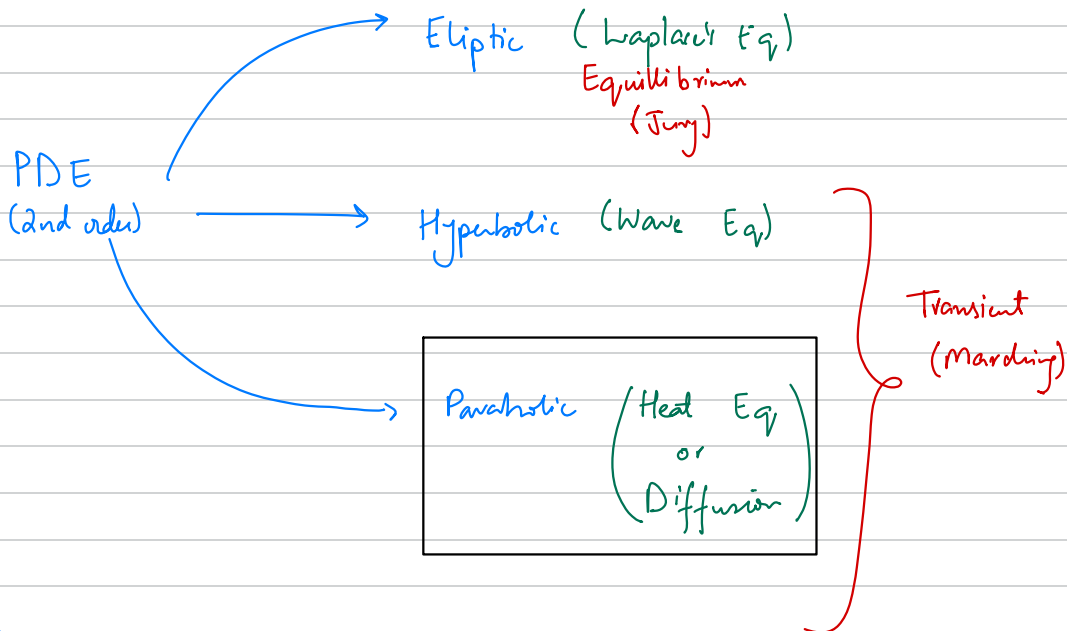
→ Laplace's Equation = $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \phi = 0$

$\nabla^2 \rightarrow \text{Laplacian}$

→ Wave Equation = $\frac{\partial^2 \phi}{\partial t^2} = c \frac{\partial^2 \phi}{\partial x^2}$

→ Heat Equation = $\frac{\partial \phi}{\partial t} = K \frac{\partial^2 \phi}{\partial x^2}$

Solving PDE of order 2



Assumption $\Rightarrow T(x,y) = u(x)v(y)$
 $\hookrightarrow$ separation

$$\begin{aligned} T(x, 0) &= 0 \\ T(0, y) &= 0 \\ T(1, y) &= 0 \\ T(x, 1) &= 0 \end{aligned}$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) T = 0$$

ES 446

- Jupyter Notebook
- Solving Heat & Laplace equation with some boundary conditions.
- We don't care where these come from.
 - ↳ empirical observation
 - ↳ Derivation from fundamental laws } ways to find!
- We only care solving, that too subject to Boundary conditions

$$\frac{d^2 f}{dx^2} = 0$$

$$\text{given } f(0) = 1$$

$$\therefore \frac{df}{dx} = C_1$$

$$\therefore C_1 x + C_2 = 1$$

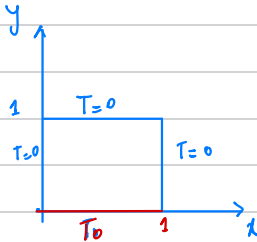
$$f = C_1 x + C_2$$

$$C_2 = 1$$

$$\therefore \boxed{f = C_1 x + 1}$$

now we need another condition

- Need to implement this in python & then plot!



$$\nabla^2 T(x,y) = 0$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) T(x,y) = 0$$

$$0 \leq x, y \leq 1$$

$$T(0,y) = 0$$

$$T(1,y) = 0$$

$$T(x,1) = 0$$

$$T(x,0) = T_0$$

How to find solution of a PDE?

Separation of Variables:

$$T(x,y) = u(x)v(y)$$

$$\text{eg: } e^{x+y} = \underbrace{e^x}_u \underbrace{e^y}_v$$

} we'll not cover every solution in this solution, but a wide range of solutions!

$$\begin{array}{c} e^{x_{(1)}} + e^{y_{(1)}} = u(x)v(y) \\ \swarrow \quad \searrow \\ u_1(x)v_1(y) \end{array}$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) (u(x)v(y)) = 0$$

$$\left(v(y) \frac{\partial^2}{\partial x^2} u(x) + u(x) \frac{\partial^2}{\partial y^2} v(y) \right) \frac{1}{u(x)v(y)} = 0$$

$$\underbrace{\frac{1}{u(x)} \frac{\partial^2}{\partial x^2} u(x)}_{-\alpha^2} + \underbrace{\frac{1}{v(y)} \frac{\partial^2}{\partial y^2} v(y)}_{\alpha^2} = 0$$

$$-\alpha^2 = \frac{1}{u(x)} \frac{\partial^2}{\partial x^2} u(x) \quad \alpha^2 = \frac{1}{v(y)} \frac{\partial^2}{\partial y^2} v(y)$$

} partial derivatives can now be made ordinary derivatives.

$$-\alpha^2 = \frac{1}{u(x)} \frac{d^2 u(x)}{dx^2} \quad \alpha^2 = -\frac{1}{v(y)} \frac{d^2 v(y)}{dy^2}$$

$$\frac{d^2}{dx^2} u(x) = -\alpha^2 u(x)$$

$$\frac{d^2}{dy^2} v(y) = \alpha^2 v(y)$$

$\swarrow$
 $\sin(x)$
 $\cos(x)$
 $-\sin(x)$

if $\sin(\alpha x)$ then: $\sin(\alpha x)$
 $\alpha \cos(\alpha x)$
 $-\alpha^2 \sin(\alpha x)$

or we could: $\sin(\alpha x + \beta) \approx A \sin(\alpha x + \beta)$

$$\therefore u(x) = A \sin(\alpha x + \beta)$$

$$T(0, y) = u(0) v(y) = 0$$

$$T(1, y) = u(1) v(y) = 0$$

This means $u(0) = 0$ } $v(y) \neq 0$, $\therefore$ then
 This means $u(1) = 0$ } $T(x, y)$ would always
 be zero!

$$u(0) = 0 = A \sin(0 + \beta), \quad \beta = 0, \pi, 2\pi, \dots$$

$$u(1) = 0 = A \sin(\alpha(1) + 0) = 0$$

$$A \sin(\alpha(0)) = 0$$

$$\therefore \alpha = n\pi$$

$$n = 0, 1, 2, 3, 4, \dots$$

$$\therefore u(x) = A \sin(n\pi x)$$

$$\alpha^2 v(y) = \frac{d^2}{dy^2} v(y)$$

$$\therefore v(y) = \boxed{e^{\alpha y}, e^{-\alpha y}}$$

$\alpha e^{\alpha y}$ $-\alpha e^{-\alpha y}$
 $\alpha^2 e^{\alpha y}$ $\alpha^2 e^{-\alpha y}$

Also $v(y) = \cosh$ $\cosh = \frac{e^x + e^{-x}}{2}$
 $\searrow \sinh$ sum of exp is
 $\searrow \cosh$ also a exp.

$$V(y) = A_2 \sinh(\alpha y + \beta)$$

$$T(x, y) = u(x)v(y)$$

$$\hookrightarrow A_n \sin(n\pi x) \sinh(n\pi(y-1)) \rightarrow \text{Generalize}$$

$$\sum_n A_n \sin(n\pi x) \sinh(n\pi(y-1))$$

$$A_1 \sin(\pi x) \sinh(1\pi(y-1))$$

+

$$A_2 \sin(2\pi x) \sinh(2\pi(y-1))$$

+

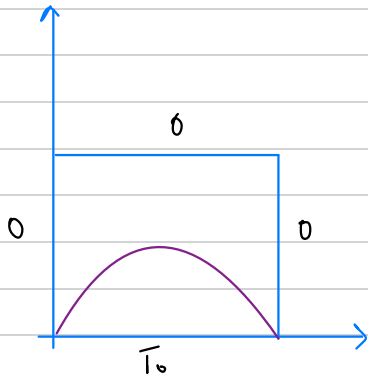
$$A_3 \sin(3\pi x) \sinh(3\pi(y-1))$$

and so on.

Lecture 3 (week 2) Monday 26/01/26

~~Hangar Air~~

Apparently araan bein small :)



diffusion, analogous to R in v-se

$$0 = D \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) T(x, y, t)$$

hyperbolic trigonometric functions don't oscillate

→ Solve $\nabla \sin(n\pi x) \sin(n\pi (y-1))$

→ generalised sol $\Rightarrow \sum_{n=1}^{\infty} A_n \sin(n\pi x) \sinh(n\pi (y-1))$
 $T(x, 0) = T_0$

$$A_n = \frac{2 T_0 ((-1)^n - 1)}{n \pi \sinh(n\pi)}$$

→ BVPs limit us to physical boundaries.

→ T_{avg} = stationary = time independent.

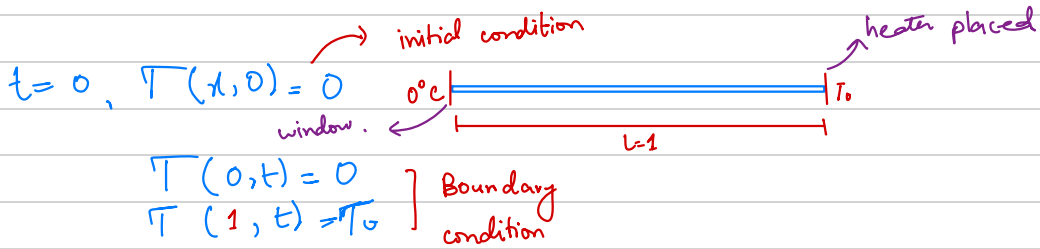
→ IVP : initial value problems

→ Marching = Propagation = Transient = time dependent.

↳ Can have both IVP & BVP

↳ you are to march outwards.

→ Determine the Transient temperature in a 1D solid with thermal diffusivity α if the initial temp is 0° & if at all subsequent times, the temperature of the left side is held at 0° while the right side is held at T_0



$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$ → need to solve this through the above conditions.

One part of the func is time ind $= u(x)$
& one will be time dep $= v(x,t)$

steady glimmer
 $\uparrow$
 $T = u(x) + v(x, t)$ $\rightarrow$ karmamey ka glimmer!

$\rightarrow$ After an infinitely long time we will have only the part of the T working where the "x" part will dominate

$\rightarrow v(x, t)$ will essentially be the part immediately after the heater is turned on because there is a more fluctuating increase in the temperature of the room.

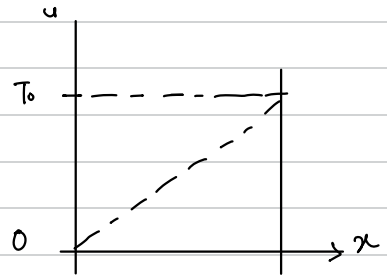
$\frac{d^2 u}{dx^2} = 0$ with BC $u(0) = 0$
 $u(1) = T_0$
 $\rightarrow u = c_1 x + c_2$

$u(0) = 0$ (left side of rod)

$c_1(0) + c_2 = 0, \quad c_2 = 0$

$c_1(1) + 0 = T_0 \quad c_1 = T_0$

$\therefore u = T_0 x + 0$



$\frac{\partial v}{\partial t} = \alpha \frac{\partial^2 v}{\partial x^2}$

with BC: $v(0, t) = v(1, t) = 0$
 $v(x, 0) = -T_0 x$

$v(x, t) = V(t) X(x)$